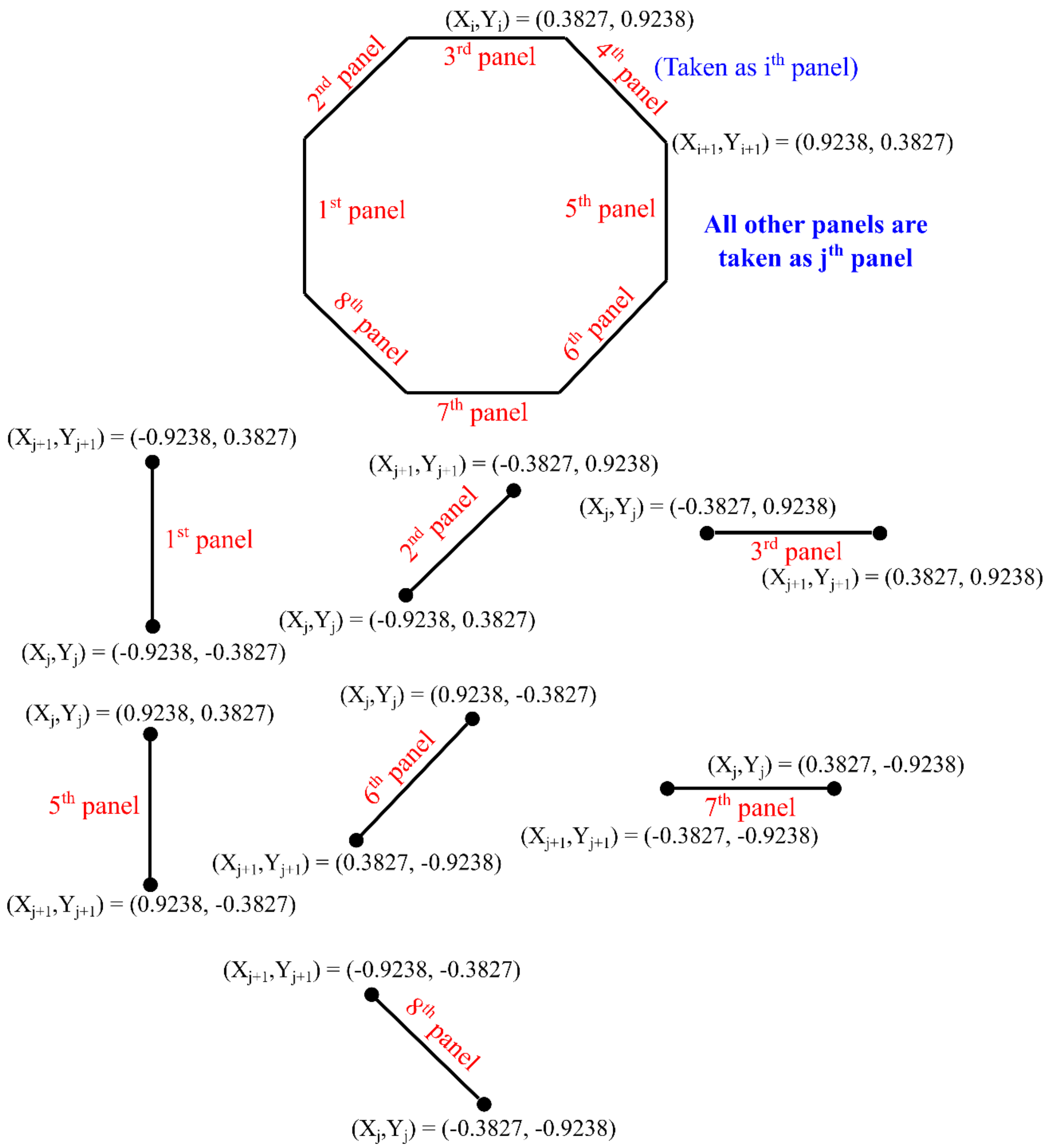

Evaluation of I for all the panels (keeping the 4th panel as the ith panel):

$$A = -(x_i - X_j)\cos\Phi_j - (y_i - Y_j)\sin\Phi_j$$

$$B = (x_i - X_j)^2 + (y_i - Y_j)^2$$

$$C = \sin(\Phi_i - \Phi_j)$$

$$D = (y_i - Y_j)\cos\Phi_i - (x_i - X_j)\sin\Phi_i$$

$$\bar{S}_j = \sqrt{(X_{j+1} - X_j)^2 + (Y_{j+1} - Y_j)^2}$$

$$E = \sqrt{B - A^2} = (x_i - X_j)\sin\Phi_j - (y_i - Y_j)\cos\Phi_j$$

From any standard table of integrals:

$$I_{i,j} = \frac{C}{2}\ln\left(\frac{\bar{S}_j^2 + 2A\bar{S}_j + B}{B}\right) + \frac{D - AC}{E}\left(\tan^{-1}\left(\frac{\bar{S}_j + A}{E}\right) - \tan^{-1}\left(\frac{A}{E}\right)\right)$$ **(Keep in mind that you need to change the settings of the calculator to Radians while calculating I_{ij}).**

Phi value for all the j panels:

$\phi_j\left(1^{st}\text{ panel}\right) = 90^\circ$

$\phi_j\left(2^{nd}\text{ panel}\right) = 45^\circ$

$\phi_j\left(3^{rd}\text{ panel}\right) = 0^\circ$

$\phi_j\left(5^{th}\text{ panel}\right) = 270^\circ$

$\phi_j\left(6^{th}\text{ panel}\right) = 225^\circ$

$\phi_j\left(7^{th}\text{ panel}\right) = 180^\circ$

$\phi_j\left(8^{th}\text{ panel}\right) = 135^\circ$

1^{st} panel as j^{th} panel and 4^{th} panel as i^{th} panel:

$$\phi_i = 315^\circ;\ \phi_j = 90^\circ$$

Convert the phi to radians:

$$\phi_i = 1.5708\text{ (radians)};\ \phi_j = 5.4978\text{ (radians)}$$

From the 4^{th} panel (i.e. i^{th} panel)

$$x_i = x_{control} = \frac{0.3827 + 0.9238}{2} = 0.6532$$

$$y_i = y_{control} = \frac{0.9238 + 0.3827}{2} = 0.6532$$

Length of the panel, $\bar{S}_j = 0.7652$

A = -1.0360; B = 3.5603; C = -0.7071; D = 1.8477; E = 1.5770

$$I_{4,1} = \frac{C}{2}\ln\left(\frac{\bar{S}_j^2 + 2A\bar{S}_j + B}{B}\right) + \frac{D - AC}{E}\left(\tan^{-1}\left(\frac{\bar{S}_j + A}{E}\right) - \tan^{-1}\left(\frac{A}{E}\right)\right)$$

$$\text{Term 1} = \frac{C}{2}\ln\left(\frac{\bar{S}_j^2 + 2A\bar{S}_j + B}{B}\right) = \frac{-0.7071}{2}\ln\left(\frac{0.7652^2 + 2(-1.0360)(0.7652) + 3.5603}{3.5603}\right)$$

$$\text{Term 1} = 0.1166$$

$$\text{Term 2} = \frac{D - AC}{E}\left(\tan^{-1}\left(\frac{\bar{S}_j + A}{E}\right) - \tan^{-1}\left(\frac{A}{E}\right)\right)$$

$$\text{Term 2} = 0.2908$$

$$I_{4,1} = \text{Term 1} + \text{Term 2} = 0.1166 + 0.2908 = 0.4073$$

<u>3rd panel as jth panel and 4th panel as ith panel:</u>

$$\phi_i = 315^\circ;\ \phi_j = 0^\circ$$

Convert the phi to radians:

$$\phi_i = 1.5708 \text{ (radians)};\ \phi_j = 0 \text{ (radians)}$$

From the 4th panel (i.e. ith panel)

$$x_i = x_{control} = \frac{0.3827 + 0.9238}{2} = 0.6532$$

$$y_i = y_{control} = \frac{0.9238 + 0.3827}{2} = 0.6532$$

Length of the panel, $\bar{S}_j = 0.7652$

A = -1.0360; B = 1.1464; C = -0.7071; D = 0.5412; E = 0.2705

$$I_{4,3} = \text{Term 1} + \text{Term 2} = 0.7274 - 0.3745 = 0.3529$$

<u>5th panel as jth panel and 4th panel as ith panel:</u>

$$\phi_i = 315^\circ;\ \phi_j = 270^\circ$$

Convert the phi to radians:

$$\phi_i = 1.5708 \text{ (radians)}; \ \phi_j = 4.7124 \text{ (radians)}$$

From the 4th panel (i.e. ith panel)

$$x_i = x_{control} = \frac{0.3827 + 0.9238}{2} = 0.6532$$

$$y_i = y_{control} = \frac{0.9238 + 0.3827}{2} = 0.6532$$

Length of the panel, $\overline{S}_j = 0.7652$

A = 0.2705; B = 0.1464; C = 0.7071; D = 0; E = 0.2706

$$I_{4,5} = \text{Term 1} + \text{Term 2} = 0.7275 - 0.3747 = 0.3528$$

<u>6th panel as jth panel and 4th panel as ith panel:</u>

$$\phi_i = 315^\circ; \ \phi_j = 225^\circ$$

Convert the phi to radians:

$$\phi_i = 1.5708 \text{ (radians)}; \ \phi_j = 3.9270 \text{ (radians)}$$

From the 4th panel (i.e. ith panel)

$$x_i = x_{control} = \frac{0.3827 + 0.9238}{2} = 0.6532$$

$$y_i = y_{control} = \frac{0.9238 + 0.3827}{2} = 0.6532$$

Length of the panel, $\overline{S}_j = 0.7652$

A = 0.5412; B = 1.1464; C = 1.0000; D = 0.5412; E = 0.9238

$$I_{4,6} = \text{Term 1} + \text{Term 2} = 0.4018 - 0 = 0.4018$$

<u>7th panel as jth panel and 4th panel as ith panel:</u>

$$\phi_i = 315^\circ;\ \phi_j = 180^\circ$$

Convert the phi to radians:

$$\phi_i = 1.5708 \text{ (radians)};\ \phi_j = 3.1416 \text{ (radians)}$$

From the 4th panel (i.e. ith panel)

$$x_i = x_{control} = \frac{0.3827 + 0.9238}{2} = 0.6532$$

$$y_i = y_{control} = \frac{0.9238 + 0.3827}{2} = 0.6532$$

Length of the panel, $\overline{S}_j = 0.7652$

A = 0.2705; B = 2.5603; C = 0.7071; D = 1.3065; E = 1.5770

$$I_{4,6} = \text{Term } 1 + \text{Term } 2 = 0.1165 + 0.2908 = 0.4073$$

<u>8th panel as jth panel and 4th panel as ith panel:</u>

$$\phi_i = 315^\circ;\ \phi_j = 135^\circ$$

Convert the phi to radians:

$$\phi_i = 1.5708 \text{ (radians)};\ \phi_j = 2.3562 \text{ (radians)}$$

From the 4th panel (i.e. ith panel)

$$x_i = x_{control} = \frac{0.3827 + 0.9238}{2} = 0.6532$$

$$y_i = y_{control} = \frac{0.9238 + 0.3827}{2} = 0.6532$$

Length of the panel, $\overline{S}_j = 0.7652$

A = -0.3826; B = 3.5603; C = 0; D = 1.8477; E = 1.8477

$$I_{4,6} = \text{Term } 1 + \text{Term } 2 = 0 + 0.4084 = 0.4084$$